

```
import 'package:flutter/material.dart';
void main()
{
  runApp(Myapplication());
}
class Myapplication extends StatelessWidget{
  @override
Widget build(BuildContext context)
{
  return MaterialApp(
    home: Scaffold(
      appBar: AppBar(
        title: const Text('First Flutter Application'),
      ), )) ;
}
}
```

First Flutter Application

DEBUG

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApplication());
}

class MyApplication extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('TIC TAC TOE'),
        ),
        body: Center(
          child: Text(
            "Let's play!",
            style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
          ),
        ),
      ),
    );
  }
}
```

first flutter application

DEBUG

Hello

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('drawer holder'),
        ),
        drawer: Drawer(
          child: ListView(
            padding: EdgeInsets.zero,
            children: <Widget>[
              DrawerHeader(
                decoration: BoxDecoration(
                  color: Colors.blue,
                ),
                child: Text(
                  'Drawer Header',
                  style: TextStyle(
                    color: Colors.white,
                    fontSize: 24,
                  ),
                ),
              ),
              ListTile(
                leading: Icon(Icons.home),
                title: Text('Home'),
                onTap: () => Navigator.pop(context),
              ),
              ListTile(
                leading: Icon(Icons.settings),
                title: Text('Settings'),
                onTap: () => Navigator.pop(context),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

```
    ),  
    body: Center(  
      child: Text('Drawer Example App'),  
    ),  
  ),  
);  
}  
}
```

Drawer Header



Home



Settings

DEBUG

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('drawer holder'),
        ),
        drawer: Drawer(
          child: ListView(
            padding: EdgeInsets.zero,
            children: <Widget>[
              DrawerHeader(
                decoration: BoxDecoration(
                  color: Colors.blue,
                ),
                child: Text(
                  'Drawer Header',
                  style: TextStyle(
                    color: Colors.white,
                    fontSize: 24,
                  ),
                ),
              ),
              ListTile(
                leading: Icon(Icons.home),
                title: Text('Home'),
                onTap: () => Navigator.pop(context),
              ),
              ListTile(
                leading: Icon(Icons.settings),
                title: Text('Settings'),
                onTap: () => Navigator.pop(context),
              ),
              ListTile(
                leading: Icon(Icons.logout),
```



```
        title: Text('logout'),
        onTap: () => Navigator.pop(context),
      ),
    ],
  ),
),
body: Center(
  child: Text('Drawer Example App'),
),
),
);
}
}
```

Drawer Header

-  Home
-  Settings
-  logout

DEBUG

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Floating Action Button Example'),
        ),
        body: Center(
          child: Text('Press the button below!'),
        ),
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            // Action to perform when the button is pressed
            print('Button Pressed!');
          },
          child: Icon(Icons.add),
          backgroundColor: Colors.blue,
        ),
      ),
    );
  }
}
```

Floating Action Button Example

DEBUG

Press the button below!



```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Floating Action Button Example'),
        ),
        body: Center(
          child: Text('Press the button below!'),
        ),
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            // Action to perform when the button is pressed
            print('Button Pressed!');
          },
          child: Icon(Icons.search),
          backgroundColor: Colors.blue,
        ),
      ),
    );
  }
}
```

Floating Action Button Example

DEBUG

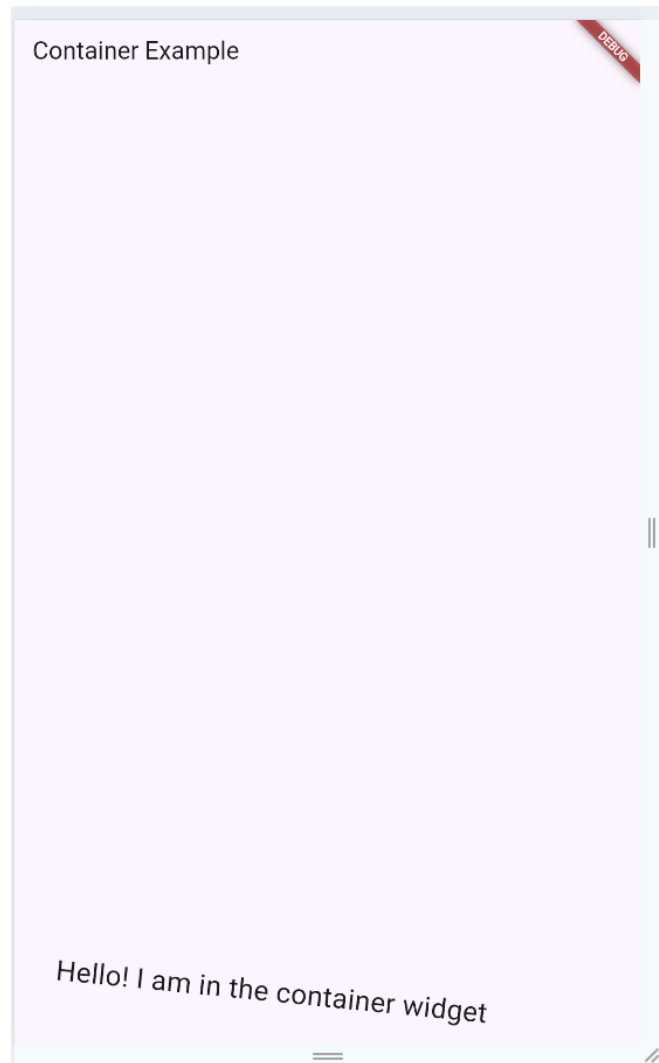
Press the button below!



```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Container Example'),
        ),
        body: Center(
          child: Container(
            padding: EdgeInsets.all(35),
            margin: EdgeInsets.all(20),
            alignment: Alignment.bottomRight,
            transform: Matrix4.rotationZ(0.1),
            child: Text(
              "Hello! I am in the container widget",
              style: TextStyle(fontSize: 25),
            ),
          ),
        ),
      ),
    );
  }
}
```



```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Constraints Example'),
        ),
        body: Center(
          child: Container(
            constraints: BoxConstraints(
```



```

        minWidth: 100,
        maxWidth: 200,
        minHeight: 100,
        maxHeight: 200,
    ),
    color: Colors.blue,
    child: Center(
      child: Text(
        'Constrained Container',
        style: TextStyle(
          color: Colors.white,
          fontSize: 18,
        ),
      ),
    ),
  ),
),
);
}
}

```

Constraints Example

DEBUG

Constrained Container

EXERCISE:1

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(title: Text('Row and Column Example')),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Text('This is a Column', style: TextStyle(fontSize: 22)),

              SizedBox(height: 20), // spacing

              Row(
                mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                children: [
                  Container(
                    color: Colors.red,
                    width: 60,
                    height: 60,
                  ),
                  Container(
                    color: Colors.green,
                    width: 60,
                    height: 60,
                  ),
                  Container(
                    color: Colors.blue,
                    width: 60,
                    height: 60,
                  ),
                ],
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

```
    ),  
  
    SizedBox(height: 20),  
  
    Text('End of Example', style: TextStyle(fontSize: 18)),  
  ],  
),  
,  
,  
);  
}  
}}
```

Row and Column Example

DEBUG

This is a Column



End of Example

EXERCISE:2

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Column Example',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('Ramaiah Chairman Profile'),
        ),
        body: const Center(
          child: UserProfile(),
        ),
      ),
    );
  }
}

class UserProfile extends StatelessWidget {
  const UserProfile({super.key});

  @override
  Widget build(BuildContext context) {
    return const Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: <Widget>[
        CircleAvatar(
          radius: 50,
          backgroundImage: NetworkImage(
            'https://tse4.mm.bing.net/th?id=OIP.Qg7TYqt7lUzqcou6WlYYIwAAAA&pid=Api&P=0&h=220'), // Replace with your image URL
        ),
      ],
    );
  }
}
```

```
    ),
    SizedBox(height: 20),
    Text(
      'Dr. M.S.Ramaiah',
      style: TextStyle(
        fontSize: 24,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
  SizedBox(height: 10),
  Text(
    'Founder Chairman',
    style: TextStyle(
      fontSize: 18,
      color: Colors.grey,
    ),
  ),
),
SizedBox(height: 20),
Padding(
  padding: EdgeInsets.symmetric(horizontal: 20),
  child: Text(
    'Mathikere Sampangi Ramaiah was born on 20 April 1922, in
Madhugiri to Sampangappa and Narasamma.',
    textAlign: TextAlign.center,
    style: TextStyle(fontSize: 16),
  ),
),
],
);
}
```

Ramaiah Chairman Profile

DEBUG



Dr. M.S.Ramaiah

Founder Chairman

Mathikere Sampangi Ramaiah was born on 20 April 1922, in Madhugiri to Sampangappa and Narasamma.

EXERCISE:3

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter TextButton Example',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('TextButton Example'),
        ),
        body: Center(
          child: MyTextButton(),
        ),
      ),
    );
  }
}

class MyTextButton extends StatelessWidget {
  const MyTextButton({super.key});

  @override
  Widget build(BuildContext context) {
    return TextButton(
      onPressed: () {
        print('Button Pressed!');
      },
      child: const Text(
        'Click Me',
        style: TextStyle(fontSize: 18),
      ),
    ),
  }
}
```

```
        style: ButtonStyle(  
            foregroundColor: MaterialStateProperty.all<Color>(Colors.blue), //  
Text color  
            overlayColor:  
MaterialStateProperty.all<Color>(Colors.blueAccent.withOpacity(0.2)), //  
Color when pressed  
        ),  
    );  
}  
}
```

TextButton Example

DEBUG

Click Me

```
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:9101?uri=http://127.0.0.1:45775/m9xQpWUjfqY=
Button Pressed!
Button Pressed!
Button Pressed!
```

EXERCISE:4

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter ElevatedButton Example',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('ElevatedButton Example'),
        ),
        body: Center(
          child: MyElevatedButton(),
        ),
      ),
    );
  }
}

class MyElevatedButton extends StatelessWidget {
  const MyElevatedButton({super.key});

  @override
  Widget build(BuildContext context) {
    return ElevatedButton(
      onPressed: () {
        print('Button Pressed!');
      },
      child: const Text(
```

```
        'Click Me',
        style: TextStyle(fontSize: 18),
    ),
    style: ElevatedButton.styleFrom(
        backgroundColor: Colors.blue,    // Background color
        foregroundColor: Colors.white,   // Text color
        shadowColor: Colors.blueAccent,
        elevation: 5,
        padding: const EdgeInsets.symmetric(horizontal: 24, vertical: 12),
    ),
);
}
}
```

ElevatedButton Example

DEBUG

Click Me

A Dart VM Service on Chrome is available at: <http://127.0.0.1:45217/dL5LVxTehdA=>
The Flutter DevTools debugger and profiler on Chrome is available at: <http://127.0.0.1:9102?uri=http://127.0.0.1:45217/dL5LVxTehdA=>
Button Pressed!
Button Pressed!

EXERCISE:5

```
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter OutlinedButton Example',
      home: Scaffold(
        appBar: AppBar(
          title: const Text('OutlinedButton Example'),
        ),
        body: const Center(
          child: MyOutlinedButton(),
        ),
      ),
    );
  }
}

class MyOutlinedButton extends StatelessWidget {
  const MyOutlinedButton({super.key});

  @override
  Widget build(BuildContext context) {
    return OutlinedButton(
      onPressed: () {
        // This function is called when the button is pressed.
        // It prints a message to the debug console.
        debugPrint('Button Clicked!');
      },
    );
  }
}
```

```
child: const Text(  
  'Click Me',  
  style: TextStyle(fontSize: 18),  
) ,  
style: OutlinedButton.styleFrom(  
  foregroundColor: Colors.blue, // Text color  
  side: const BorderSide(color: Colors.blue), // Border color  
  padding: const EdgeInsets.all(16),  
  shape: RoundedRectangleBorder(  
    borderRadius: BorderRadius.circular(8),  
  ),  
) ,  
);  
}  
}
```

A Dart VM Service on Chrome is available at: <http://127.0.0.1:37733/0ubzvS5Bl0g=>
The Flutter DevTools debugger and profiler on Chrome is available at:
<http://127.0.0.1:9103?uri=http://127.0.0.1:37733/0ubzvS5Bl0g=>
Button Clicked!
Button Clicked!

OutlinedButton Example

DEBUG

Click Me

FLUTTER ALERT DIALOG

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  void _showAlertDialog(BuildContext context) {
    showDialog(
      context: context,
      builder: (BuildContext context) {
        return AlertDialog(
          title: Text("Alert Dialog"),
          content: Text("This is a basic alert dialog."),
          actions: <Widget>[
            TextButton(
              child: Text("Close"),
              onPressed: () {
                Navigator.of(context).pop();
              },
            ),
          ],
        );
      },
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
```

```
        title: Text("Flutter Alert Dialog Example"),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
            _showAlertDialog(context);
          },
          child: Text("Show Alert Dialog"),
        ),
      ),
    );
  }
}
```

Flutter Alert Dialog Example

DEBUG

Show Alert Dialog

ICON DESIGN

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Flutter Icons Example"),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: <Widget>[
            Icon(
              Icons.star,
              size: 100.0,
              color: Colors.yellow,
            ),
            Text("Star Icon", style: TextStyle(fontSize: 20.0)),
            SizedBox(height: 20.0),
            Icon(
              Icons.favorite,
              size: 100.0,
              color: Colors.red,
            ),
            Text("Favorite Icon", style: TextStyle(fontSize: 20.0)),
            SizedBox(height: 20.0),
          ],
        ),
      ),
    );
  }
}
```

```
        Icon(  
            Icons.local_florist,  
            size: 100.0,  
            color: Colors.green,  
        ),  
        Text("Florist Icon", style: TextStyle(fontSize: 20.0)),  
    ],  
),  
),  
);  
}  
}
```

Flutter Icons Example

DEBUG



Star Icon



Favorite Icon



Florist Icon

FLUTTER LIST

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  final List<String> items = [
    'Item 1',
    'Item 2',
    'Item 3',
    'Item 4',
    'Item 5',
  ];

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Flutter List Example"),
      ),
      body: ListView.builder(
        itemCount: items.length,
        itemBuilder: (BuildContext context, int index) {
          return ListTile(
            title: Text(items[index]),
          );
        },
      ),
    );
  }
}
```


}

Flutter List Example

DEBUG

Item 1

Item 2

Item 3

Item 4

Item 5

FLUTTER TOAST

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  void showToast(BuildContext context) {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(
        content: Text('Hello, Flutter Toast!'),
        duration: Duration(seconds: 2),
      ),
    );
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter Toast Example'),
      ),
      body: Center(
        child: ElevatedButton(
          onPressed: () => showToast(context),
          child: Text('Show Toast'),
        ),
      ),
    );
  }
}
```

```
);  
}  
}
```

Flutter Toast Example

DEBUG

Show Toast

FLUTTER THEME

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        scaffoldBackgroundColor: Colors.black, // Set background color to
black
        primaryColor: Colors.yellow, // Set text color to yellow
      ),
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter Theme Example'),
      ),
      body: Center(
        child: Text(
          'Welcome to Flutter Themes',
          style: TextStyle(
            fontSize: 24,
            color: Colors.yellow, // Set text color explicitly
          ),
        ),
      ),
    );
  }
}
```

}

Welcome to Flutter Themes